

# COMP232 Lab 4

## Diffie-Hellman Key Exchange

This lab session focuses on the Diffie-Hellman key exchange protocol. Download the following from the Canvas page (use `DH.py` and `DH3.py` for Python):

- `DHKeyAgreement2.java`
- `DHKeyAgreement3.java`

### References:

- [JCA/JCE Reference Manual](#)
- [Python Diffie-Hellman key exchange Reference Manual](#)
- [Python AES Reference Manual](#)

## 1. Two-party Diffie-Hellman key exchange

Compile and run `DHKeyAgreement2.java` (`DH.py` for Python).

Look into the code and see how the key exchange is implemented and how the shared secret is further used in the AES encryption.

Then try the following:

- Find a key size setting in the code. It should be 2048 bits. Try to increase it to 4096, compile and run again. What do you observe? (Hint: It should not work, the compiler will report a possible range of key sizes).
- Try to set the key size to the minimal possible value, compile and run the program again. Try other possible values. Notice how the size of the shared secret changes.
- Measure the time (see Lab 3 for the details) required to perform key generation and key agreement for the minimal and maximal possible key sizes.

## 2. Three-party Diffie-Hellman key exchange

- Compile, and run `DHKeyAgreement3.java` (`DH3.py` for Python).
- Look into the code and understand how the key exchange is implemented for three parties.
- Write the process of key exchange for three parties in terms of the modular exponent, similar to the two-party protocol that we discussed in Lecture 6.